

Enclosure Specification

CJ2 Voice Emergency Checklist Unit

Cessna Citation CJ2 (CE 525A) — bench / training prototype

Two-piece housing: DZUS-rail panel bezel + remote processor box

Mounting, materials, audio, thermal, environmental targets,
labeling, deliverables, and reference dimensions.

DEMO / TRAINING USE ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS

Not a certified avionics installation. Borrows airworthiness conventions (DZUS, DO-160) so the prototype fits and looks right, but it is a mock-up. Real installation requires a licensed A&P/IA and FAA approval (337/STC/TSO).

1 Design Intent & Architecture

This document is written for the mechanical / industrial-design engineer who will model, prototype, and fabricate the housing for the voice emergency-checklist unit. It assumes the electronics, pin-out, and behavior are fixed and documented in the companion `TECH_DATA.md` and the wiring diagram (reproduced in §9).

The unit splits into two physical pieces joined by a short harness, because the pilot-facing controls and the processor have very different placement needs:

Piece	Contents	Where it lives	Why
A. Panel bezel (pilot-facing)	Split-legend annunciator switch, PTT button (optional)	Front instrument panel / pedestal, on the DZUS rail	Must be seen and reached by the crew; the dark-cockpit annunciator must sit in the pilot's normal scan
B. Remote processor box (blind)	ESP32-S3 board, isolated audio-input stage (isolation transformer + I2S codec ADC / analog tap), isolated audio-output stage (line driver/DAC + output isolation transformer → COM3-style channel), two microSD breakouts (Config Card slot 1 + Data Card slot 2), lamp-driver board, power conditioning. (INMP441 mic + MAX98357A speaker-amp: bench-test only, not installed.)	Avionics bay / behind-panel shelf, blind-mounted	Keeps heat, the SD slots, and wiring out of the panel; only the bezel needs panel real estate

A single all-in-one panel box is acceptable for a pure bench demo, but the two-piece split is the recommended target: it mirrors how real remote-mount avionics are installed and routes audio-panel signals (receive tap + COM3 output) cleanly from the avionics bay.

PANEL (crew side)	AVIONICS BAY (blind)
+-----+	+-----+
[VOICE CHKLST OFF] bezel harness	ESP32-S3
[VOICE CHKLST FALT] <=====>	isolated audio-IN stage
o PTT (optional) (D-sub or	isolated audio-OUT stage
+-----+ circular)	Config Card (front access)
Piece A	Data Card (front access)
	lamp-driver board
	power conditioning
	+-----+
	Piece B

2 Piece A — Panel Bezel

2.1 Form factor & mounting

- Mounting standard: DZUS rail (the CJ2 pedestal and most of its panel use DZUS). Fastener pitch (hole-to-hole) = 3/8 in (9.525 mm); fastener clearance hole 0.255 in (6.48 mm).
- Bezel height must be a whole multiple of 3/8 in (the “DZUS rhythm”). Target a 3-unit panel = 1.125 in (28.575 mm) tall (switch + PTT fit comfortably; no speaker/grille cutout required). 4-unit = 1.5 in (38.1 mm) if additional connector cutouts are needed.
- Standard pedestal panel width = 5.75 in (146.05 mm) nominal aluminum ($\approx$ 144.45 mm usable face). Use this width so the bezel drops into a standard slot.
- Backplate: 1/16 in (1.6 mm) aluminum (6061-T6) per DZUS convention; first / last fastener centers $1.5 \times 3/8$ in = 14.29 mm from each end.
- Alternative (round-hole) mount: if no DZUS slot is available, provide a variant that fits a standard 3-1/8 in (79.4 mm) instrument cutout with four 6-32 screws on the standard bolt circle, so it can replace a blanking plate.

2.2 Front-face layout (top → bottom, dark-cockpit order)

- Split-legend annunciator switch (Applied Avionics VIVISUN / Korry). Cutout per the specific switch series' datasheet — confirm the exact part before cutting. A typical VIVISUN 1-pole rectangular bezel is $\approx 15 \times 15$ mm to 19×19 mm face; provide snap or screw retention per that series.
- Legend engraving: top half = VOICE CHKLST OFF (white); bottom half = VOICE CHKLST FAULT (amber). Legend orientation must read upright when panel-installed.
- PTT button (optional) — momentary, guarded or recessed so it is not bumped. If the aircraft already has a yoke / PTT tie-in, omit this and route PTT through the harness.
- No speaker grille. The installed configuration has no onboard speaker; checklist audio is delivered in-headset via the isolated COM3-style output channel. No acoustic aperture is required on the bezel for speaker output.

2.3 Bezel material & finish

- Face: 6061-T6 aluminum, 2.0–3.0 mm, or ABS/PC for a non-structural demo.
- Finish: matte black, low-gloss (≤ 10 gloss units) to suppress windshield glare — standard for instrument-panel parts and consistent with dark-cockpit philosophy.
- Engraving: laser-etched + paint-fill in white and amber to match the legend colors, OR rely solely on the switch's internal legend (preferred — fewer painted parts to chip).
- Edges chamfered 0.5 mm; no sharp corners facing the crew.

3 Piece B — Remote Processor Box

3.1 Size & internal layout

- Size the internal volume around the largest board path = the ESP32-S3 DevKitC-1 ($\approx 70 \times 26$ mm) plus the isolated audio-output stage board (line driver/DAC + output isolation transformer), the audio-input stage board (isolation transformer + codec), two microSD breakouts (Config Card slot 1 + Data Card slot 2), and the 2-channel lamp-driver board. Practical outer envelope: $\approx 110 \times 80 \times 45$ mm (L $\times$ W $\times$ H). Confirm against the actual stacked board set.
- Use internal standoffs / PCB rails (M2.5 brass inserts) so boards are screwed down, not floating — important for the vibration environment (§6).
- The installed audio input is the audio-panel tap (isolated analog line tap or buffered digital I2S), not an onboard microphone. No acoustic port is needed on the remote box for the installed configuration. The audio-panel tap takes already-mixed crew audio from the panel, eliminating sensitivity to cockpit fan or avionics noise entirely.
- No onboard speaker or speaker grille in the installed build. The output goes out through the isolated line-level stage to the panel's COM3-style channel; the crew hears checklist audio in-headset, not from a speaker in the box.

3.2 Access & connectors

- microSD access — two front-accessible slots: both the Config Card (slot 1) and Data Card (slot 2) must be externally accessible (push-push carriers or slots on a removable face) so either card can be swapped without opening the sealed box. Label: CONFIG CARD (SL0T 1) — FAT32 and DATA CARD (SL0T 2) — FAT32.
- USB-C service port: a covered / recessed USB-C pass-through for flashing and power on the bench. Add a silicone plug or hinged cover; it is not for in-service use.
- Main connector (bezel harness): one circular bayonet connector (e.g. a small MIL-style / M12) OR a D-sub carrying: bezel switch SELECT, the two legend-lamp drives, PTT, and power / ground. Pin-out to be finalized from board_pins.h. Use a keyed, positive-latching connector — no loose flying leads.
- AUDIO IN connector (isolated input tap): a separate panel connector / feedthrough for the galvanically-isolated receive-only audio-panel INPUT tap line ($600\ \Omega$ line level or buffered I2S). Label: AUDIO IN — PANEL TAP (ISOLATED).
- AUDIO OUT connector (isolated COM3 output): a separate panel connector / feedthrough for the galvanically-isolated line-level OUTPUT to the dedicated COM3-style audio-panel channel. Label: AUDIO OUT — COM3 ISOLATED. Separate connector from the input tap; keep the TX and RX paths on distinct connectors.
- Power: accept bench supply (USB 5 V $\geq$ 1 A). If a 28 VDC aircraft-bus mock-up is wanted, include a 28 V $\rightarrow$ 5 V DC-DC ($\geq$ 2 A) inside the box with input transient protection (TVS + fuse) — but mark clearly that this is a demo regulator, not DO-160 qualified.

3.3 Box material & EMI

- Material: aluminum (extruded or machined) preferred for the remote box — it doubles as an EMI shield and a heatsink. ABS/PC acceptable for a pure desk demo.

- If plastic, add a conductive shield liner or coating (nickel paint) tied to ground, since the unit sits near sensitive avionics; good practice even for a mock-up.
- Single-point chassis ground stud bonded to the connector shell and the ESP32 ground.

4 Audio Interface — Isolated Panel Connectors

The installed configuration has no onboard speaker and no speaker grille. Checklist audio is delivered in-headset via two electrically separate, galvanically-isolated audio-panel connections on two dedicated external connectors (see §3.2):

- AUDIO IN — PANEL TAP (ISOLATED): galvanically-isolated receive-only tap of the aircraft audio panel (analog isolated line tap via 600 Ω :600 Ω aviation isolation transformer, e.g. Allen Avionics AGL series, OR buffered receive-only digital I2S, selectable per install). No signal path back toward the panel from this connector. Use a shielded audio cable; keep the run to the panel short and away from ignition wiring.
- AUDIO OUT — COM3 ISOLATED: galvanically-isolated line-level TX to the dedicated COM3-style audio-panel input channel. Output impedance: 600 Ω nominal (per panel COM3 input spec). The isolation transformer on the TX line is the primary galvanic barrier; a device fault cannot key, jam, load, or back-feed the panel's other channels or required COM radios. Use shielded cable; confirm output level is set conservatively so COM3 advisory audio cannot mask required ATC audio on other channels.
- No acoustic grille needed on either the bezel or the box for the installed configuration. Remove grille-pattern cutouts from the CAD model; plug any bench-prototype holes cleanly.

5 Thermal

- The ESP32-S3 running ESP-SR is the main heat source but is low power (a few hundred mW typical); no fan is required.
- Provide passive convection: a few vent slots low and high on the remote box, or bond the ESP module's ground plane to the aluminum wall with a thermal pad.
- Do not seal the box fully airtight if relying on convection. If sealing is required for dust / water, switch to conduction cooling (thermal pad to the aluminum case) and verify the internal rise stays < 20 °C above ambient at 55 °C ambient.
- Keep the DC-DC (if fitted) on its own thermal path; it is the second heat source.

6 Environmental Targets

DO-160G categories a real unit in this location would target. For the prototype, treat these as design guidance, not test obligations.

DO-160G section	Category to design toward	Note for this unit
§4 Temperature / Altitude	Cat A2 (controlled cockpit)	-15 °C to +55 °C operating; pressurized cabin
§7 Shock & crash safety	Operational shock + crash-safety	Mounts / standoffs must retain boards; nothing becomes a projectile
§8 Vibration	Cat S (fixed-wing)	Locking hardware; conformal-coat the boards; no press-fit-only parts
§6 Humidity	Cat A	48-hr; conformal coat recommended
§16/17 Power input / spike	—	Add TVS + fuse on any 28 V input mock-up
§20/21 RF emission / susceptibility	—	Aluminum case + grounded shield; Wi-Fi stays disabled (it is, in firmware)
§25 ESD	—	Bond exposed metal; recessed connectors
§26 Flammability	—	Use flame-rated plastics (UL94 V-0) if not aluminum

The firmware already runs fully offline with Wi-Fi and Bluetooth off, which materially helps the RF-emission story for any future qualification.

7 Labeling & Markings

- Required placard on the bezel and the box: DEMO / TRAINING ONLY — NOT FOR FLIGHT.
- Box exterior: unit name, a serial / asset field, the config-card format (FAT32), and the USB service-port “bench use only” note.
- Annunciator legends: VOICE CHKLST OFF (white, top), VOICE CHKLST FAULT (amber, bottom).
- Use amber for caution / white for status, consistent with FAA AC 25-11B color conventions used elsewhere in this project.

8 Deliverables Requested from the Engineer

- 1 3D CAD (STEP + native) of both pieces, with the board set and switch modeled in place.
- 2 2D drawings with the DZUS hole pattern, the switch cutout (dimensioned from the chosen switch datasheet), AUDIO IN / AUDIO OUT connector cutouts, and any remaining panel apertures — fully dimensioned, GD&T where it matters (switch cutout, DZUS holes). No speaker-grille pattern required.
- 3 Connector pin-out mapping the bezel-to-box harness to board_pins.h signals.
- 4 A bench-printable prototype (FDM/SLA) of both pieces for fit-check before any aluminum.
- 5 Mechanical BOM (fasteners, inserts, connectors, AUDIO IN / AUDIO OUT panel feedthroughs, gaskets, conformal-coat).

- 6 Tolerance call-outs: switch cutout ± 0.1 mm; DZUS holes ± 0.1 mm on the 9.525 mm pitch; general ± 0.25 mm.

8.1 Open items for the engineer to confirm

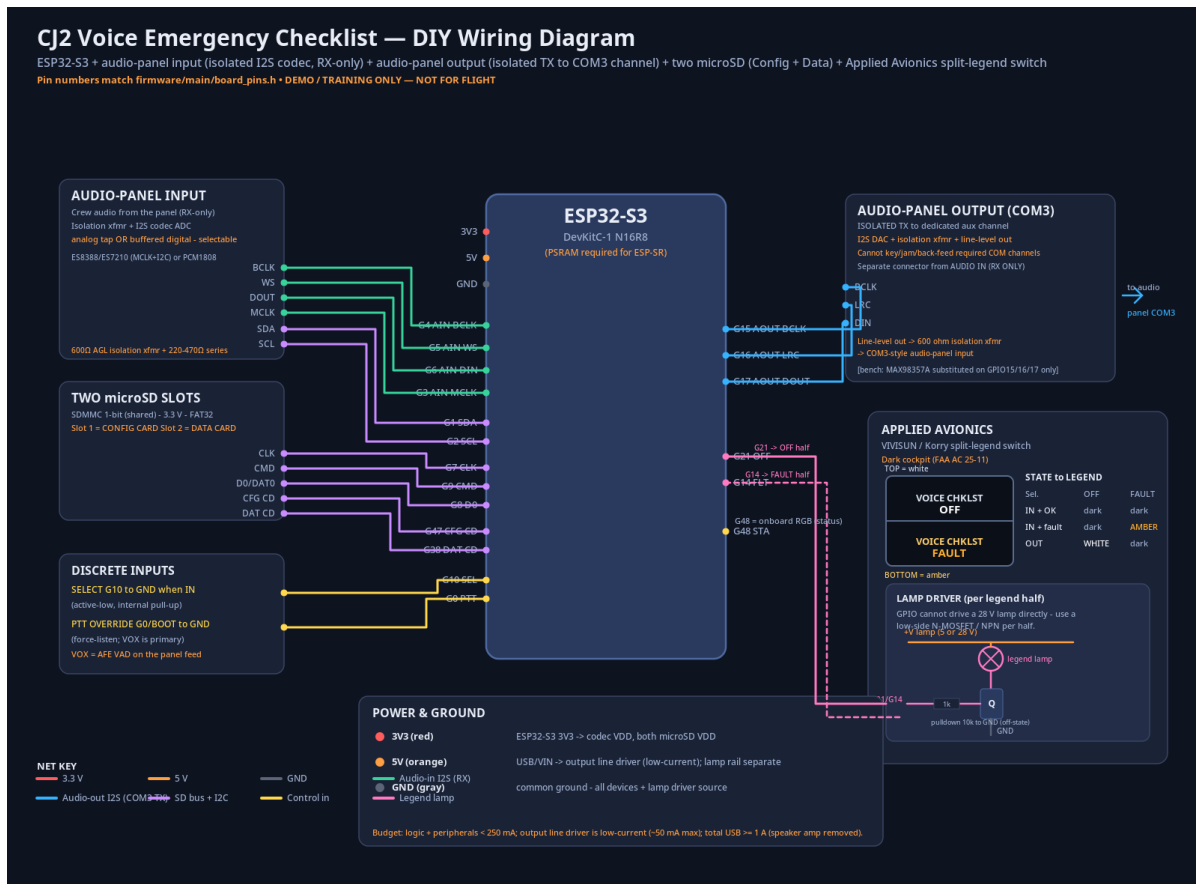
The single most critical unknown is the exact Applied Avionics switch part number — it drives the bezel cutout and depth. Nothing else can be finalized until it is fixed.

- Mounting reality in the target panel: is there a free DZUS slot, or must this go into a 3-1/8 in round hole or a custom sub-panel? Pick the §2.1 variant accordingly.
- Audio-panel connection type: confirm whether the install uses the analog isolated line tap or the buffered digital I2S path — this drives the audio-input stage BOM and the AUDIO IN connector pin-out. The Config Card sets the active source.
- COM3 channel input impedance — confirm the panel's COM3-style input spec so the isolated output stage (line driver + transformer) can be matched to the correct output level and impedance (typically 600 Ω).
- Whether a 28 V input is wanted, or USB 5 V bench power is sufficient for the demo.

9 Reference Dimensions & Wiring Context

Item	Value	Source
DZUS fastener pitch	3/8 in = 9.525 mm	DZUS rail standard
DZUS clearance hole	0.255 in = 6.48 mm	DZUS rail standard
Panel backplate thickness	1/16 in = 1.6 mm	DZUS convention
First fastener offset from edge	1.5 × pitch = 14.29 mm	DZUS convention
Standard pedestal panel width	≈ 146 mm alu (144.45 mm face)	DZUS panel convention
Standard round instrument hole	3-1/8 in = 79.4 mm	Instrument panel standard
Remote box target envelope	≈ 110 × 80 × 45 mm	This design (confirm vs. boards)
Audio IN connector	600 Ω line level (isolated) or buffered I2S	TECH_DATA.md §D.3.1 / D.3.2
Audio OUT connector (COM3)	600 Ω line-level output (isolated, via transformer)	TECH_DATA.md §D.3.3a

The harness between Piece A and Piece B carries the signals shown below. The full electrical reference (pin map, per-device wiring, lamp driver) is in the companion technical packet.



System wiring diagram (from the companion technical packet).

10 Source References

#	Reference	Source
1	DZUS rail dimensions & the “DZUS rhythm”	MyCockpit panel-building guide
2	DZUS rail product reference	Dallas Avionics
3	Standard 3-1/8 in instrument cutout / adapter	Aircraft Spruce
4	DO-160G environmental categories (summary)	Wikipedia
5	Environmental qualification guidance	FAA AC 21-16G
6	Annunciator color / dark-cockpit conventions	FAA AC 25-11B
7	LED lighted pushbutton / split-legend switches	Applied Avionics (VIVISUN)

DEMO / TRAINING USE ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS. This enclosure is for a bench / training prototype. It is not a certified avionics installation and is not an STC, TSO, or field-approval package. Any installation in or near an actual aircraft requires a licensed A&P/IA and the appropriate FAA approval. Repository: github.com/flas-tech/cj2-voice-emergency-checklist (MIT License).